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Solid State Theory — Exercise 2

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Link: `see Ilias page`

1. Sommerfeld expansion

The Sommerfeld expansion allows to approximate integrals of functions with the Fermi function at low temperatures and it is given by

$$I = \int_{-\infty}^{\infty} d\varepsilon a(\varepsilon) n_F(\varepsilon - \mu) = \int_{-\infty}^{\infty} d\varepsilon A(\varepsilon)(-n'_F(\varepsilon - \mu)) = A(\mu) + \frac{\pi^2}{6} a'(\mu)(k_B T)^2 + \mathcal{O}(k_B T)^4 \quad (1)$$

with the Fermi function $n_F(E) = 1/(e^{E/(k_B T)} + 1)$ and $A'(\varepsilon) = a(\varepsilon)$ and the properties that the product $A(\varepsilon) n_F(\varepsilon - \mu)$ vanishes at $|\varepsilon| \to \infty$. It can be derived (you do not have to do this here) from (i) first performing an integration by parts, (ii) expanding the function A in a Taylor series in powers of $\varepsilon - \mu$, and (iii) finally integrating each term of the series.
Consider the free energy density of a Fermi gas

$$f(T, \mu) = -k_B T \int_{-\infty}^{\infty} d\varepsilon \nu(\varepsilon) \log\left(1 + e^{-\frac{\varepsilon - \mu}{k_B T}}\right). \quad (2)$$

with a density of states $\nu(\varepsilon)$ that is smooth close to the chemical potential μ.

a) Evaluate the behavior of the specific heat $C = -T\partial_T^2 f$ at constant μ in the limit of low temperatures. First show by explicitly evaluating the second order derivative that

$$C = k_B^2 T \int_{-\infty}^{\infty} d\varepsilon \nu(\varepsilon) \left(\frac{\varepsilon - \mu}{k_B T}\right)^2 (-n'_F(\varepsilon - \mu)). \quad (3)$$

Second, apply Eq. (1) to obtain the behavior at low T.

b) Evaluate the temperature dependence of the chemical potential μ up to order $\mathcal{O}(k_B T)^2$ assuming a constant density of fermions, $n = -\partial_\mu f$. Evaluate the prefactor of the leading T dependence for a parabolic energy dispersion $(\hbar k)^2/(2m)$ in $d = 3$ and express the result in terms of the Fermi energy ε_F.

2. Kronig–Penney model with Dirac potential

Consider the one-dimensional stationary Schrödinger equation $\left(-\frac{\hbar^2\partial_x^2}{2m} + V(x) - \varepsilon\right)\psi(x) = 0$ in the presence of a periodic potential $V(x) = V_0 \sum_{m\in\mathbb{Z}} \delta(x - ma)$ consisting of equally spaced delta functions.

a) For $x \in (-a, 0)$ the wavefunction is given by a plane wave $\psi_I(x) = Ae^{iqx} + Be^{-iqx}$ with $\varepsilon = \hbar^2 q^2/(2m)$. Use Bloch's theorem $\psi(x+a) = e^{ika}\psi(x)$ to derive the form of the wavefunction $\psi_{II}(x)$ for $x \in (0, a)$. Then use the boundary conditions at $x = 0$ in order to derive a linear set of equations that determines A and B. At δ-potential $\psi(x)$ is continuous but its derivative jumps. By integrating the Schrödinger equation around $x = 0$, derive an equation for jump $\psi'_{II}(\eta) - \psi'_I(-\eta)$

$$\psi'_{II}(\eta) - \psi'_I(-\eta) = \frac{2mV_0}{\hbar^2}\psi_I(0). \tag{4}$$

Show that this set can be solved provided that q obeys the equation

$$\cos(ka) = F_v(qa) \equiv \cos(qa) + v\frac{\sin(qa)}{qa}, \tag{5}$$

with the parameter $v = \frac{mV_0 a}{\hbar^2}$.

b) In Fig. 1, we show $F_v(qa)$ and energy $\epsilon(qa) = \epsilon_0(qa)^2$ as a function of qa with $v = 10$ and $\epsilon_0 \equiv \frac{\hbar^2}{2ma^2} = 0.1$. We want to solve Eq. (5) graphically. Color on Fig. 1 the regions where Eq. (5) has solutions. What do those regions correspond to? How can you read off energies at $k = 0$ and $k = \pi/a$? Can you sketch roughly the resulting dispersion?

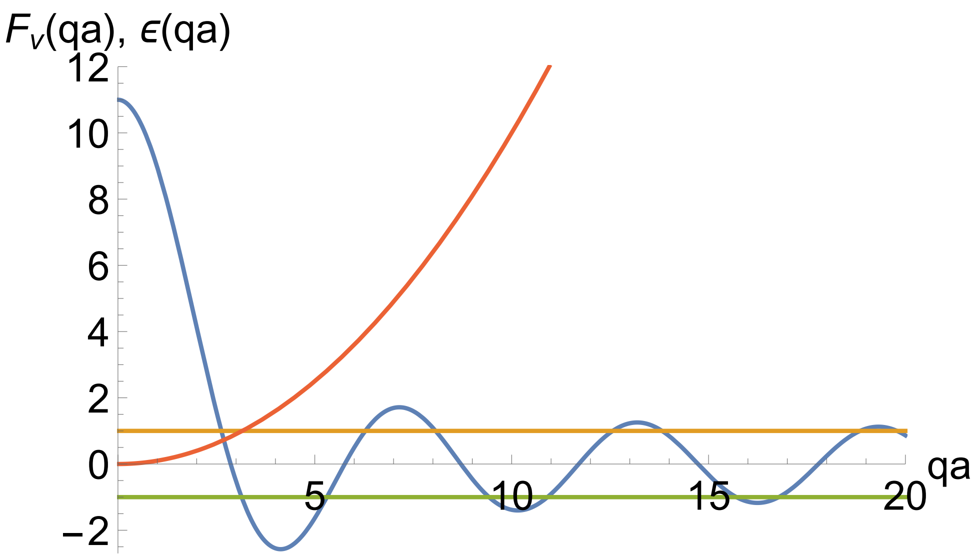

Figure 1: Function $F_v(qa)$ (blue solid line) and energy $\epsilon(qa)$ (red solid line) with $v = 10$ and $\epsilon_0 = 0.1$.

c) Consider the equation (5) in the tight-binding limit $v \gg 1$ and derive the energy dispersion $\varepsilon_{1,k}$ of the lowest band $n = 1$. First, show that for $v \to \infty$ this energy is constant and given by $\varepsilon_{1,k} \to \hbar^2\pi^2/(2ma^2)$. Second, derive the correction to $\varepsilon_{1,k}$ in first order in $1/v$ expanding q at $q_0 = \pi/a$.

3. Two-dimensional tight-binding model (C)

In the tight-binding approximation the Bloch energies $\epsilon_{n\mathbf{k}}$ of conduction electrons are obtained by diagonalizing the matrix

$$\mathcal{H}_{mm'}(\mathbf{k}) = -\sum_{\mathbf{R}} t_{m,m'}(\mathbf{R}) e^{i\mathbf{k}\cdot\mathbf{R}}, \quad m, m' = 1, 2, \ldots, N. \tag{6}$$

where N is the total number of bands and $t_{mm'}(\mathbf{R})$ is the hopping amplitude for an electron to hop between orbitals m and m' belonging to different atoms separated by the Bravais lattice vector $\mathbf{R}$. In the following, we discuss the Bloch energies and Fermi surfaces of p-electrons on a two-dimensional square lattice with lattice constant a.

a) We consider the atomic p_x and p_y orbitals ($N = 2$) with nearest-neighbor hopping t only. Why does the hopping amplitude between different orbitals vanish? The nearest-neighbour hopping amplitude among the p_x (and p_y) orbitals is much larger along the x- (and y-)direction (σ-bonding) than along the other directions (π-bonding). Taking into account only the σ-bondings, determine the Bloch energies $\varepsilon_{n\mathbf{k}}$ for the two bands $n = 1, 2$ and sketch the constant-energy surfaces. Evaluate the four points $\mathbf{k}_0$ in the first Brillouin zone where the bands are degenerate, $\varepsilon_{1,\mathbf{k}_0} = \varepsilon_{2,\mathbf{k}_0} = \varepsilon_F$ for a given Fermi energy ε_F.

b) The degeneracy of the bands at $\mathbf{k}_0$ will be lifted by including a next-nearest-neighbor hopping amplitude t' between the p_x and p_y orbitals. Why is such an amplitude t' finite? Which 2×2 matrix determines therefore the band structure? Taylor-expand $\mathcal{H}_{mm'}(\mathbf{k}_0 + \delta\mathbf{k})$ for small $\delta\mathbf{k}$ amd determine the spectrum. Can you sketch the resulting Fermi surface? This simple model explains, for example, two of the three bands observed in Sr_2RuO_4 (see Fig. 2). Which ones? Motivate the experimentally observed shape of the Fermi surface by Taylor expansion of the band dispersion around $\mathbf{k}_0$ for non-zero chemical potential.

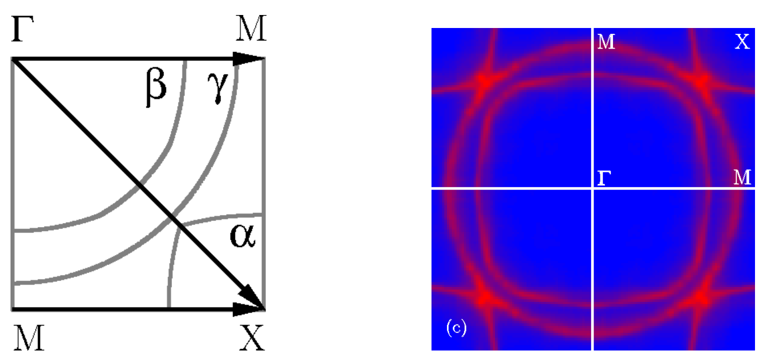

Figure 2: Theoretically (left) and experimentally (right) determined α, β and γ sheets of the Fermi surface of Sr_2RuO_4 (A. Damascelli *et al.*, Phys. Rev. Lett. **85** 5194, 2000).